

LINCOLN, NE, USA

WMO: 725510

Lat: 40.851N

Lon: 96.748W

Elev: 363

StdP: 97.04

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14939

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.1	-16.6	-23.6	0.5	-18.3	-21.2	0.6	-15.7	14.1	-2.6	12.5	-3.1	3.0	320	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	35.7	24.0	34.0	23.8	32.4	23.2	25.9	32.7	25.1	31.8	24.2	30.4	6.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.7	30.2	23.0	18.6	29.1	22.2	17.7	28.2	82.3	32.8	78.5	31.9	74.9	30.7	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	11.0	9.6	-24.0	38.7	2.5	1.6	-25.9	39.8	-27.3	40.7	-28.8	41.6	-30.6	42.7		
(5)			-24.3	27.3	2.5	0.8	-26.1	27.9	-27.5	28.3	-28.9	28.8	-30.7	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-4.0	-1.8	4.8	11.0	17.1	22.9	25.5	24.1	19.6	12.1	4.5	-2.1
	DBStd	11.59	6.64	6.84	6.68	5.44	4.77	3.75	3.33	3.25	4.70	5.34	5.57	6.03
	HDD10.0	1601	434	332	186	52	3	0	0	0	1	38	179	375
	HDD18.3	3285	692	563	420	227	82	7	1	2	41	203	415	633
	CDD10.0	2042	0	2	26	82	223	387	479	436	288	104	14	1
	CDD18.3	683	0	0	1	6	43	144	222	179	78	10	0	0
	CDH23.3	7467	0	1	21	112	499	1544	2508	1807	847	124	6	0
	CDH26.7	3191	0	0	3	30	168	638	1197	796	325	33	1	0
Wind	WSAvg	4.3	4.2	4.5	4.9	5.4	4.7	4.4	3.8	3.7	4.1	4.3	4.3	4.0
Precipitation	PrecAvg	731	15	23	51	70	104	103	76	90	90	53	33	21
	PrecMax	1050	40	78	169	183	258	328	176	218	210	137	97	87
	PrecMin	455	0	2	3	7	23	4	9	2	7	0	0	1
	PrecStd	155	10	18	41	42	47	74	38	57	56	41	29	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	20.3	25.7	30.2	33.6	36.0	38.5	37.3	34.7	30.6	23.6	15.8
		MCWB	7.4	10.6	14.9	17.5	21.0	23.6	24.4	23.8	22.5	19.7	14.2	9.7
	2%	DB	11.0	15.1	22.4	26.5	30.7	33.9	36.2	34.9	32.6	26.4	19.8	12.3
		MCWB	5.6	8.7	13.6	16.2	20.2	23.4	24.5	24.2	22.2	17.4	12.3	7.3
	5%	DB	8.0	11.5	19.0	23.6	28.4	32.1	34.3	32.9	30.5	24.0	16.6	9.5
		MCWB	4.1	6.5	11.9	14.9	18.8	22.9	24.4	23.8	21.3	16.2	10.7	5.2
	10%	DB	5.2	8.2	15.7	20.8	26.1	30.5	32.5	31.0	28.2	21.4	13.9	6.8
		MCWB	2.4	4.4	10.1	13.3	18.2	21.9	23.9	23.0	20.1	14.9	9.0	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	12.6	17.4	19.2	23.2	26.2	27.3	26.8	24.5	21.4	15.7	11.7
		MCDB	13.8	16.1	23.1	26.2	30.2	32.7	34.1	33.3	31.5	27.6	20.0	14.5
	2%	WB	6.1	9.1	15.2	17.4	21.8	24.9	26.2	25.6	23.3	19.4	13.7	7.7
		MCDB	10.4	14.9	19.0	24.2	28.1	31.5	33.2	32.2	29.9	24.1	18.3	11.5
	5%	WB	4.2	6.6	12.7	15.8	20.5	23.9	25.4	24.7	22.5	17.5	11.5	5.5
		MCDB	7.8	11.1	18.7	22.1	26.5	30.3	32.3	31.0	28.6	22.4	16.1	9.2
	10%	WB	2.5	4.6	10.2	14.1	19.0	22.9	24.6	23.8	21.4	15.6	9.1	3.6
		MCDB	5.2	8.1	15.6	20.0	24.5	29.2	31.0	29.5	26.8	20.2	13.6	6.5
Mean Daily Temperature Range	5% DB	MDBR	11.1	11.4	13.0	13.1	12.4	11.8	11.8	11.9	13.4	13.2	12.5	10.9
		MCDBR	15.8	16.9	18.3	17.7	15.5	13.5	13.9	14.2	15.1	16.7	17.3	15.6
		MCWBR	11.0	10.9	10.5	9.4	7.1	5.8	5.2	5.7	6.5	8.7	10.5	11.0
	5% WB	MCDBR	15.0	16.1	15.8	15.2	13.2	11.9	11.9	12.3	12.9	13.2	15.5	14.9
		MCWBR	10.8	10.7	9.5	9.0	6.9	6.0	5.4	5.8	6.3	7.8	10.3	10.9
Clear-Sky Solar Irradiance	taub	0.283	0.303	0.326	0.381	0.407	0.418	0.424	0.422	0.382	0.333	0.295	0.281	
	taud	2.468	2.430	2.408	2.276	2.271	2.302	2.283	2.294	2.353	2.478	2.509	2.478	
	Ebn at Noon	885	914	926	891	871	860	850	842	854	866	863	857	
	Edh at Noon	76	92	105	129	133	129	130	126	110	87	73	70	
All-Sky Solar Radiation	RadAvg	2.11	3.01	3.94	5.00	5.88	6.59	6.57	5.72	4.68	3.34	2.36	1.78	
	RadStd	0.18	0.18	0.33	0.33	0.43	0.45	0.41	0.32	0.31	0.36	0.16	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	+52
(47) Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page